

Unit 3: Introduction to Excel

❖ What is Excel?

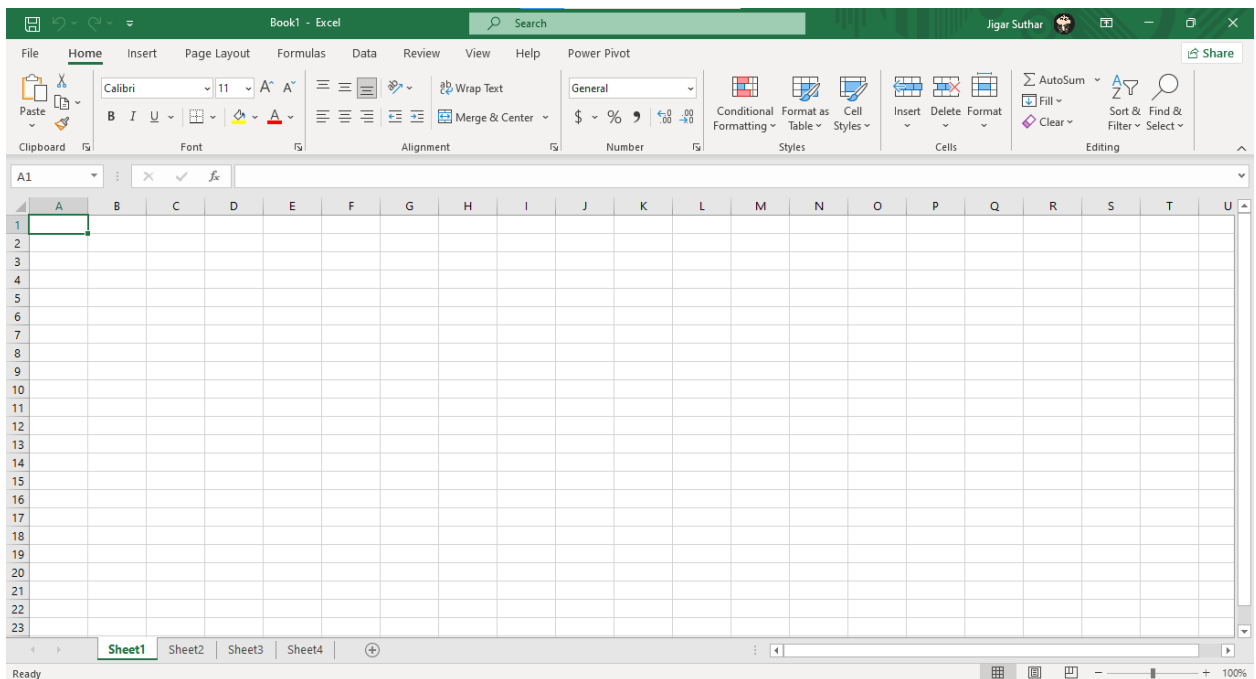
- Excel is pronounced "Eks - sel"
- It is a spreadsheet program developed by Microsoft. Excel organizes data in columns and rows and allows you to do mathematical functions. It runs on Windows, macOS, Android and iOS.
- The first version was released in 1985 and has gone through several changes over the years. However, the main functionality mostly remains the same.
- **Excel is typically used for:**
 - Analysis
 - Data entry
 - Data management
 - Accounting
 - Budgeting
 - Data analysis
 - Visuals and graphs
 - Programming
 - Financial modelling
 - And much, much more.

❖ Why use Excel?

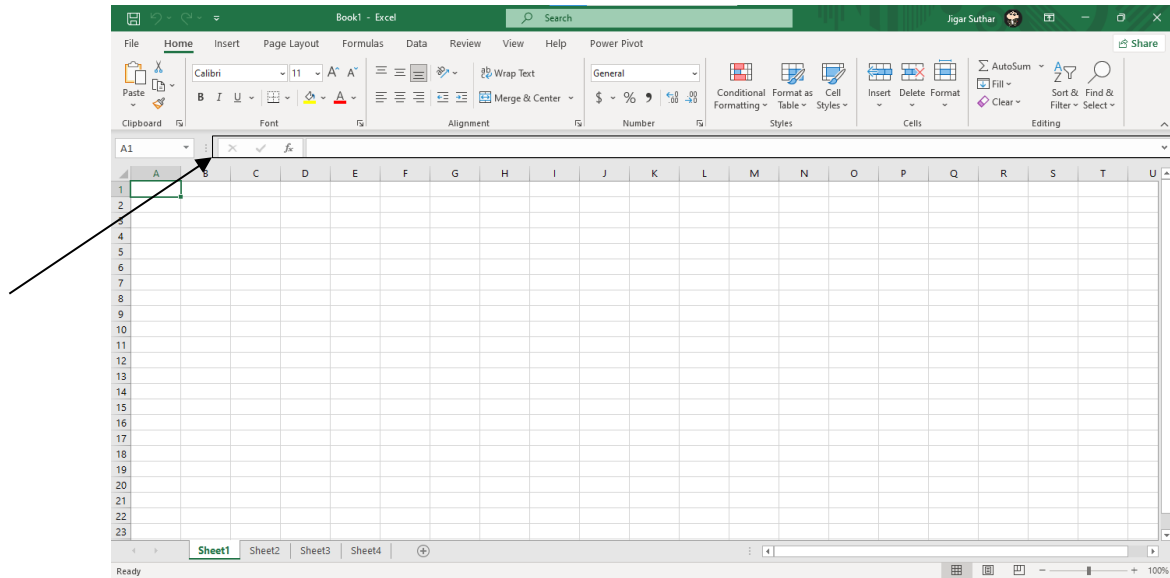
- It is the most popular spreadsheet program in the world.
- It is easy to learn and to get started.
- The skill ceiling is high, which means that you can do more advanced things as you become better.
- It can be used with both work and in everyday life, such as to create a family budget.
- It has a huge community support.
- It is continuously supported by Microsoft.

- Templates and frameworks can be reused by yourself and others, lowering creation costs.

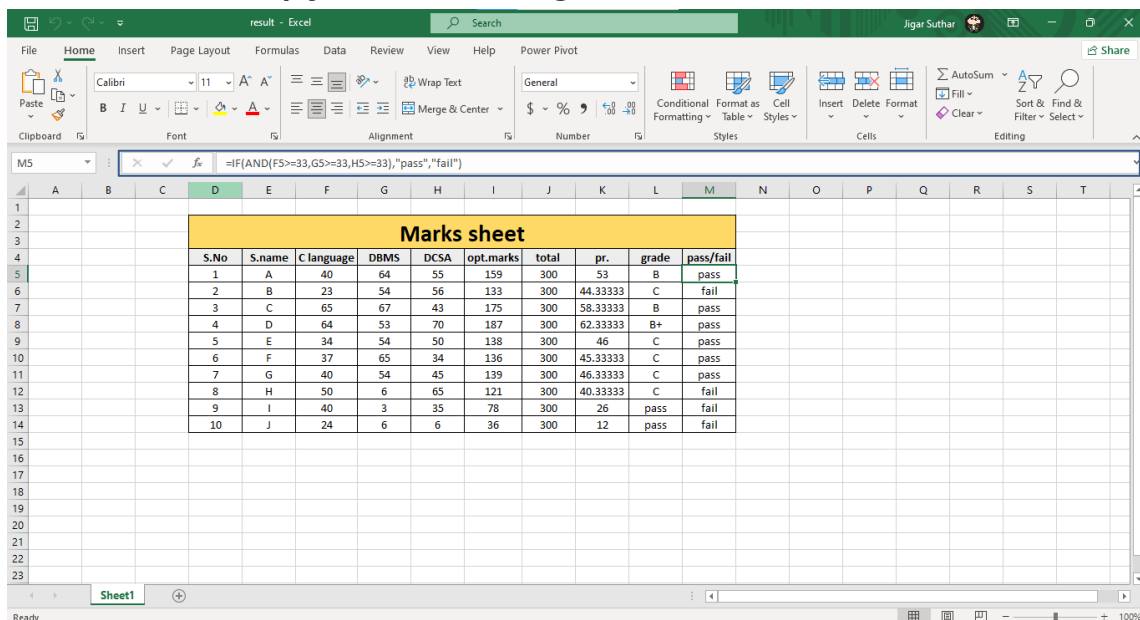
❖ **Concepts of workbook:-** In Microsoft excel, a workbook is a collection of one or more spreadsheet also called worksheet in a single file below an example of a spreadsheet called “sheet 1” in an excel workbook file called “book 1” our example also has the “sheet 1” and “sheet 3” sheet tabs which are also part of the same workbook.



❖ **Cell address:-** A cell reference, or cell address, is an alphanumeric value used to identify a specific cell in a spreadsheet. Each cell reference contains one or more letters followed by a number. The letter or letters identify the column and the number represents the row.



❖ **Formula bar:-** Excel formula bar is a special tab bar at the top of the excel worksheet window labeled with function symbol (fx) you can use it to enter a new formula or copy an existing one.



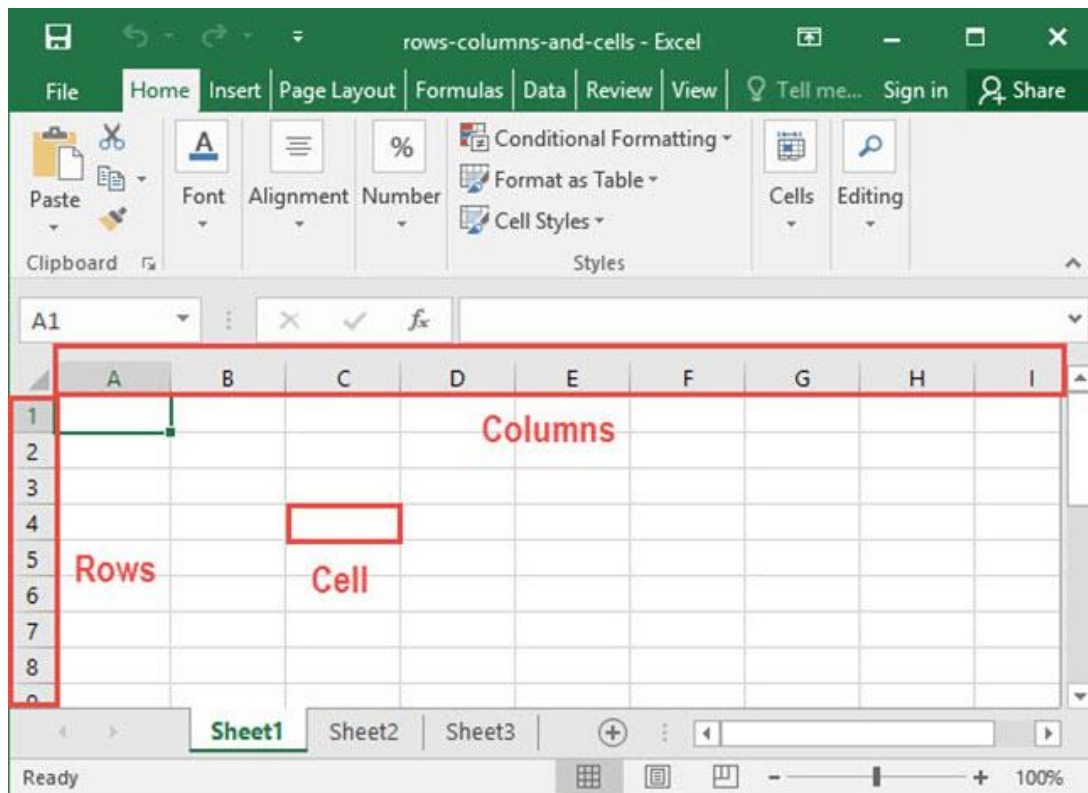
❖ **Column:-** In ms excel a column runs query in the grid layout of worksheet vertices column are numbered with alphabetic values such as A,B,C each column in

the worksheet has its own column number which is used as part of a cell.

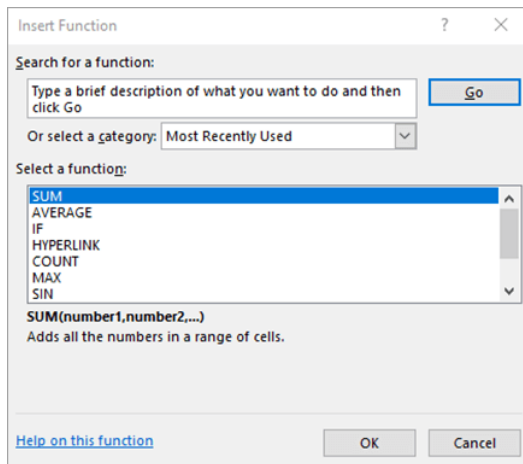
❖ **Rows:-** In ms excel a row runs horizontally in the grid layout of worksheet, horizontally rows are numbered with numeric values such as 1,2,3 each row in the worksheet has its own row number which is used as part of a cell reference such as A1,A2.

The rows themselves run horizontally on a worksheet.

❖ **Cell:-** Cell is the intersection of row and column in other words where a row and column meet, columns are identified by letters(A,B,C) while rows are identified by number (1,2,3) each has its own name or cell address based on its column and row.



❖ **Insert:-** Use the **Insert Function** dialog box to help you insert the correct formula and arguments for your needs. (To view the **Insert Function** dialog box, click )



Search for a function

Type a brief description of what you want a function to do, and then choose **Go**. A list of functions likely to fit your needs and based on your description will display in the Select a function box.

Or select a category

❖ **Delete:-** Cells can be deleted by selecting them, and pressing the delete button.

Note: The delete function will not delete the formatting of the cell, just the value inside of it.

Let's have a look at three examples.

Example 1

	A	B	C	D	E
1	Delete me				
2					
3					
4					

Pressing the delete button:

	A	B	C	D	E
1					
2					
3					
4					

Example 2

A1		f_x	=1+1			
	A	B	C	D	E	
1	2					
2						
3						
4						

Pressing the delete button:

A1	▼	f_x			
	A	B	C	D	E
1					
2					
3					
4					

Example 3

With formatting:

A1	fx	Delete me			
	A	B	C	D	E
1	Delete me				
2					
3					
4					

Pressing the delete button:

A1	fx				
	A	B	C	D	E
1					
2					
3					
4					

Note that it only deletes the value in the cells, and not the formatting (the colour).

- ❖ **Formula cell:-** In Microsoft Excel, a formula is **an expression that operates on values in a range of cells**. These formulas return a result, even when it is an error. Excel formulas enable you to perform calculations such as addition, subtraction, multiplication, and division.
- ❖ **Cell size:-** In Excel, cells are the basic unit within a spreadsheet. Each spreadsheet contains thousands of cells, which are small and rectangular boxes you see when you open a new sheet. Within these cells, you can

input the information you want to use in your spreadsheet.

Every cell is at the intersection of a row and column, and the system organizes rows by number and columns by letter. Therefore, each cell has its own cell address based on its column and row. For example, the cell in column B and row six has the cell address of B6. You can also select multiple cells at a time, which is a cell range. To refer to a cell range, list the first and last cells for that range. For example, you can write a cell range that includes the cells B3, B4, B5 and B6 as "B3:B6."

Methods for how to change cell size in Excel

Adjusting the size of a single cell typically requires adjusting all the cells in that row or column. There are several methods you can use to change the size of a cell, including the following four methods:

1. Use the Format menu

In this method, start by clicking on a cell in the row or column you wish to adjust. Next, click on the "Home" button within the top menu. From there, click on the "Cells" submenu and then "Format." A drop-down menu appears,

under which you have the option to select either "Row Height" or "Column Width." Pick the one based on which you want to adjust first.

Then, input a new number to adjust that measurement. The default row height is 15, while the default column width is 8.43. Try different adjustments to those numbers until you have your desired cell size.

Related: [Basic Excel Formulas and How To Use Them](#)

2. Use your mouse

Another simple way to adjust the column or row size of a cell is by using your mouse. Move your mouse over the boundary between one cell and the next on the edge of the sheet. For example, if you want to change the height of the first row, move your mouse between the labels "1" and "2." Similarly, to change the width of column "A," move your mouse to the boundary between the labels "A" and "B."

Once you move your mouse, the cursor changes to a black bar with two arrows pointing in opposite directions. Click your mouse and drag it to adjust the size. This can allow you to quickly adjust the height or width of an entire row or column.

3. Use AutoFit

The AutoFit feature involves setting the system to adjust cells to fit their contents automatically so that all your information is visible. To enable AutoFit, start by clicking on a cell in the row or column you want to change. Then, click on the "Home" menu button. In the "Cells" section, click on "Format." In this drop-down menu, both "AutoFit Column Width" and "AutoFit Row Height" are available. Click on the corresponding option that you want to use. This action applies the AutoFit feature to the row or column of your selected cell.

If you wish to apply this feature to an entire sheet, go to the "Edit" menu and click "Select All." Then, you can either use the "Format" menu again or double-click on a boundary. For example, double-clicking the boundary between two columns applies AutoFit to column widths, while double-clicking the boundary between two rows applies AutoFit to row heights.

Related: [Advanced Excel Skills: Definition and Examples](#)

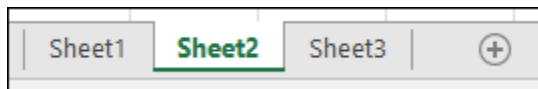
4. Merge multiple cells

If you wish to make a cell bigger without impacting the entire row or column, one option is to merge multiple cells. When you merge cells, you combine two or more cells into one that spans across multiple rows or columns. For example, merging the cells "A1" and "B1" results in a

single cell at "A1" which spreads to the boundary of "B1" and "C1," effectively removing "B1." This method allows more information to appear visible in "A1" without impacting the other cells in the "A" column.

For this method, select the cells you want to merge with your mouse. On the "Home" menu, click on "Merge & Center." This merges the cells and centers the information within the cell. If you wish to keep the current alignment, you can select the drop-down menu next to "Merge & Center" and select "Merge Cells." To undo the merging of cells, you can use this same menu and click "Unmerge Cells."

❖ **Rename sheet:-** By default, Excel names worksheets Sheet1, Sheet2, Sheet3 and so on, but you can easily rename them.



3 ways to rename a worksheet

1. Double-click the sheet tab, and type the new name.
2. Right-click the sheet tab, click **Rename**, and type the new name.

3. Use the keyboard shortcut **Alt + H > O > R**, and type the new name.

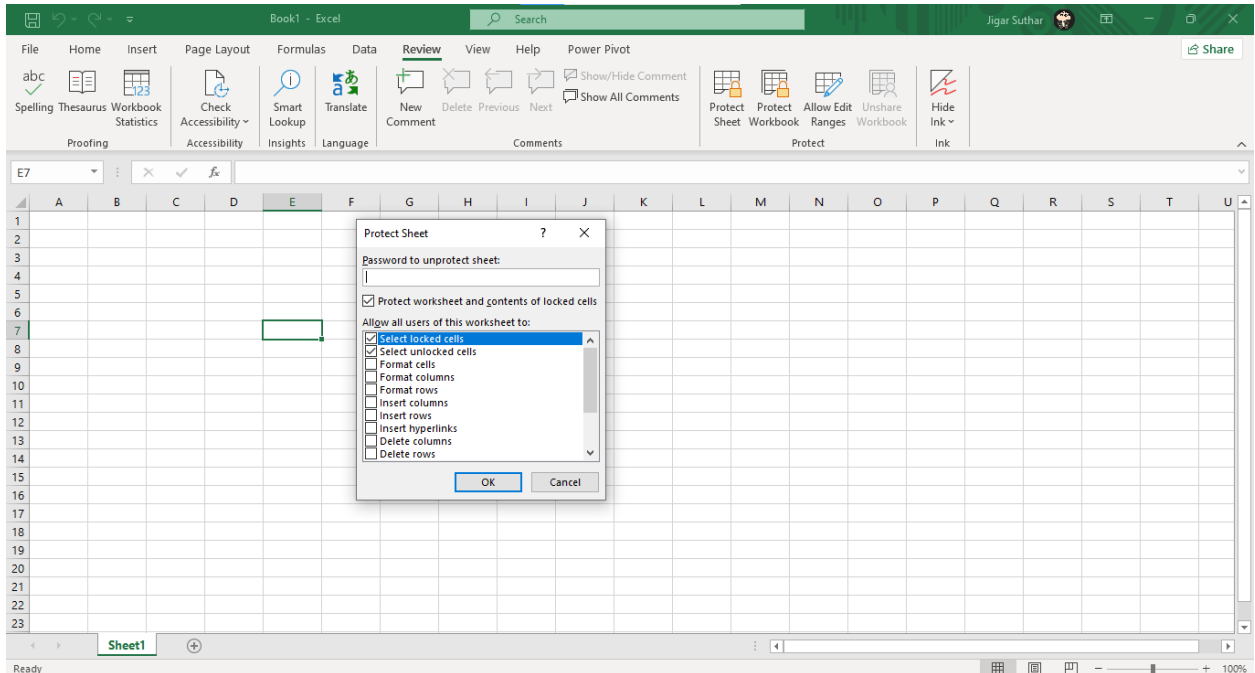
Important: Worksheet names cannot:

- Be blank .
- Contain more than 31 characters.
- Contain any of the following characters: / \ ? * : []

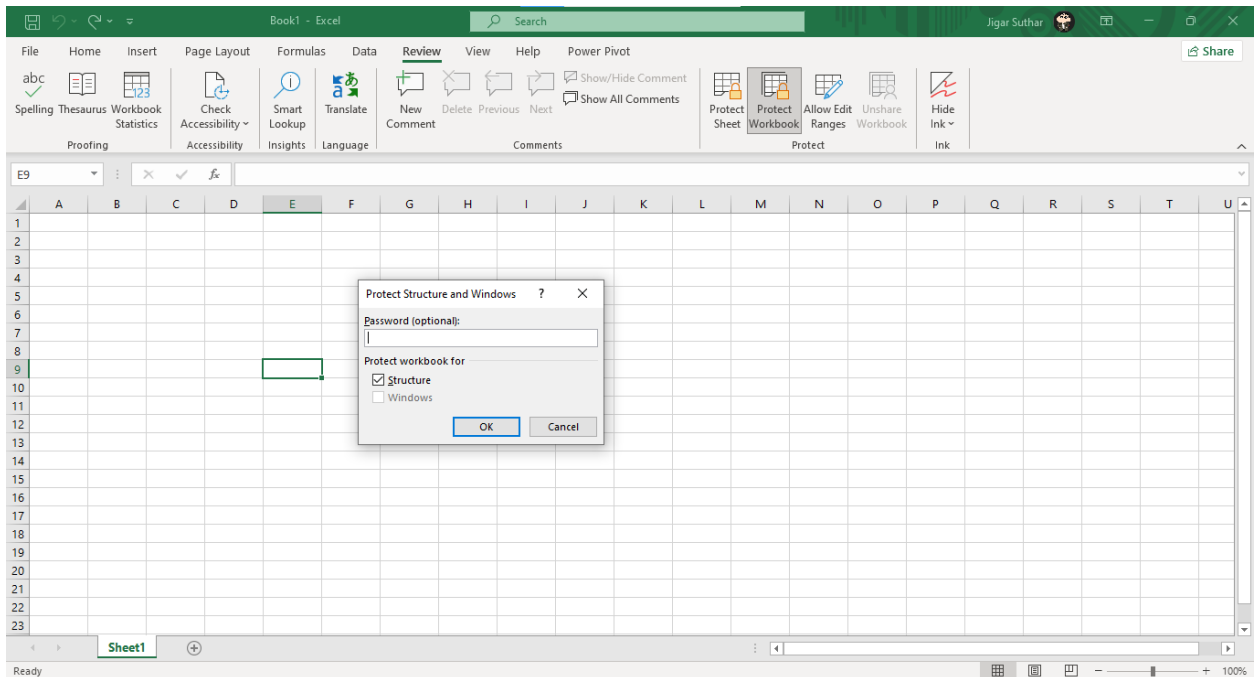
For example, 02/17/2016 would not be a valid worksheet name, but 02-17-2016 would work fine.

- Begin or end with an apostrophe ('), but they can be used in between text or numbers in a name.
- Be named "History". This is a reserved word Excel uses internally.

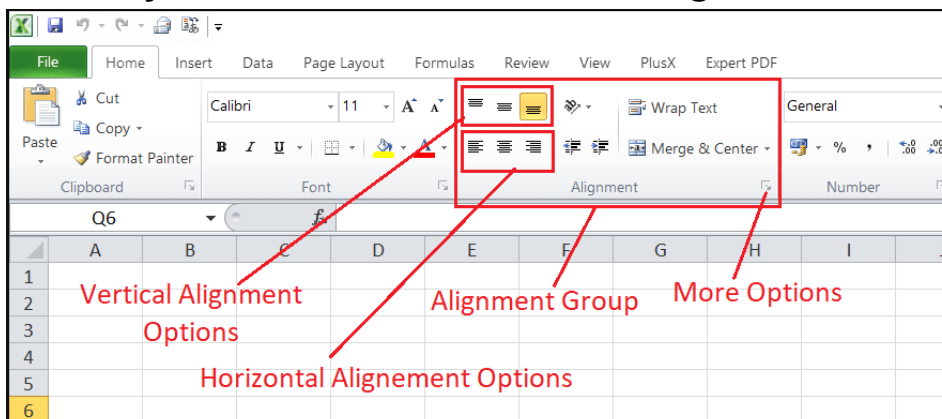
❖ **Protect sheet:-** To prevent other users from accidentally or deliberately changing, moving, or deleting data in a worksheet, you can lock the cells on your Excel worksheet and then protect the sheet with a password.



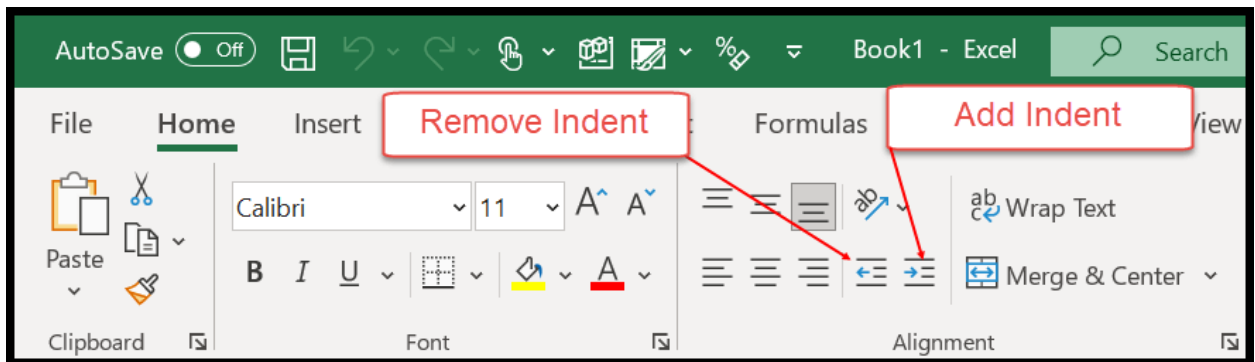
- ❖ **Lock sheet:-** Select the cells you want to take this name that the cells are ready to be locked when you protect the workbook or worksheet on the review tab in the ribbon in the changes group select either protect sheet or protect workbook and then reapply protection.



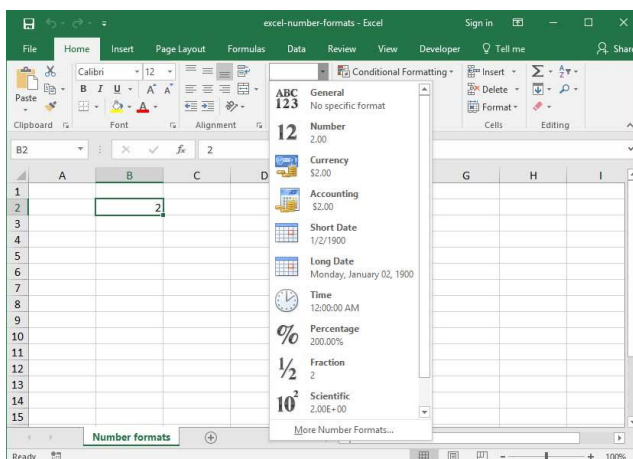
- ❖ **Alignment:-** With ms excel alignment is your tab or numbers are position in the cell you can align vertically meaning to work the top the middle or the bottom excel actually has its own default for alignment.



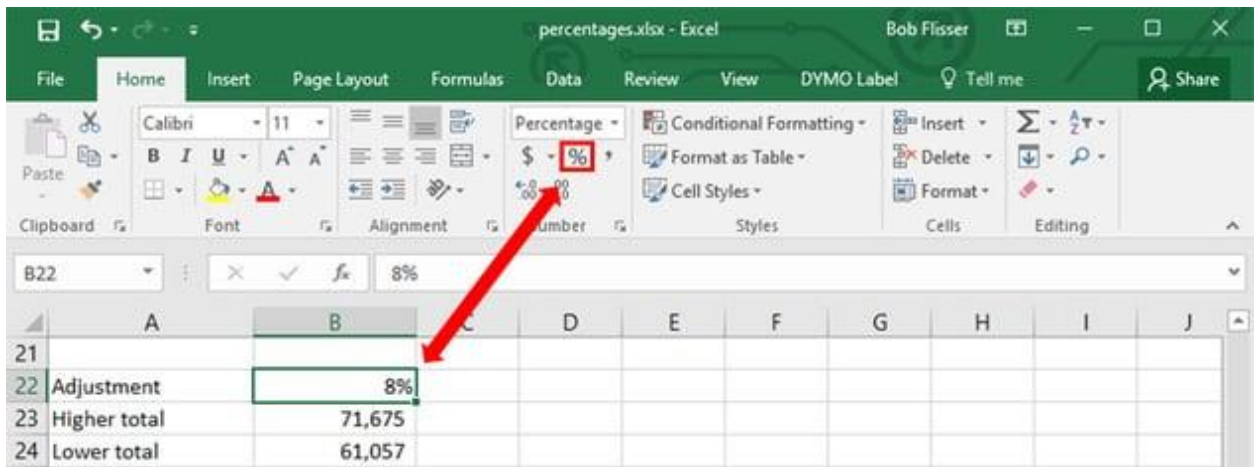
- ❖ **Indent:-** Indenting Individual Lines of Text in a Cell
Turn on "Wrap Text" using the instructions above. Click your cursor into the desired cell. Type the desired text. To go to a new line, press [Alt]+[Enter]. To indent individual lines of text, use the space bar to insert the desired space.



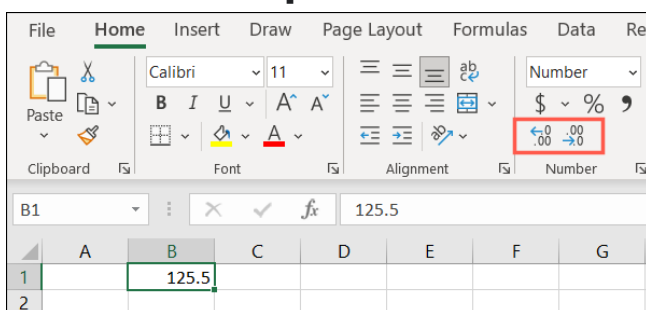
- ❖ **Number format:-** You can use number formats to change the appearance of numbers, including dates and times, without changing the actual number. The number format does not affect the cell value that Excel uses to perform calculations. The actual value is displayed in the formula bar. Excel provides several built-in number formats.



- ❖ **Percent style:-** Percentages are calculated by using the equation **amount / total = percentage**. For example, if a cell contains the formula =10/100, the result of that calculation is 0.1. If you then format 0.1 as a percentage, the number will be correctly displayed as 10%



- ❖ **Increase decimal:-** On the Home tab, click Increase Decimal or Decrease Decimal to show more or fewer digits after the decimal point.
- ❖ **Decrease decimal:-** Select the cells that you want to format. On the Home tab, click Increase Decimal or Decrease Decimal to show more or fewer digits after the decimal point.



❖ **Formulas & Charts:-**

- **Creating 2D and 3D charts (Columns, Line, Pie, Bar, Scatter):-**

To insert 3D Bar Chart (or just Bar Chart), select the data for which the Bar chart has to be generated, now **go to INSERT menu and click on 'Insert Bar Chart' command button from Charting group on the ribbon as shown below and select desired 3D chart type under '3-D Bar' heading**. A 3-D chart appears on the screen.

➤ **Column chart:-**

- **What is a Column Chart?**
- Column charts visually present data over periods of time or other category. Column charts are similar to [Bar Charts](#), but the bars on a column chart extend vertically instead of horizontally.
- A column chart is especially useful for displaying and comparing worksheet data that is arranged in columns and rows.
- A simple column chart graphs a single data series—one row or column of data. But most Excel users create charts that contain multiple data series. Our multiple data series is outlined in the worksheet below.

	A	B	C	D	E	F
1		1st Qtr	2nd Qtr	3rd Qtr	4th Qtr	Year
2	Flowers	\$220	\$260	\$270	\$130	\$880
3	Shrubs	\$200	\$280	\$290	\$180	\$950
4	Trees	\$260	\$350	\$330	\$250	\$1,190

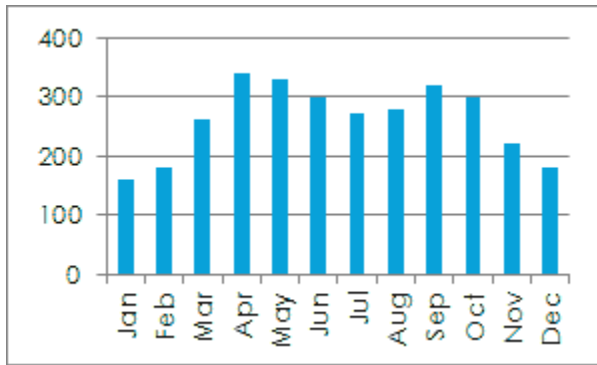
➤ Chart Data -

Multiple Data Series

- In a column chart, the vertical axis (Y-axis) typically displays numeric values and the horizontal axis (X-axis) displays time or other category. Excel plots whichever has the most entries, row or column data, on the horizontal axis.
- For example, if plotting five rows and two columns of data, the row headings would reside along the horizontal axis. This can be flipped around by pressing the **Switch Row/Column** button in the **Data** section of the ribbon's **Design** tab ([Customizing the Excel Chart](#)).

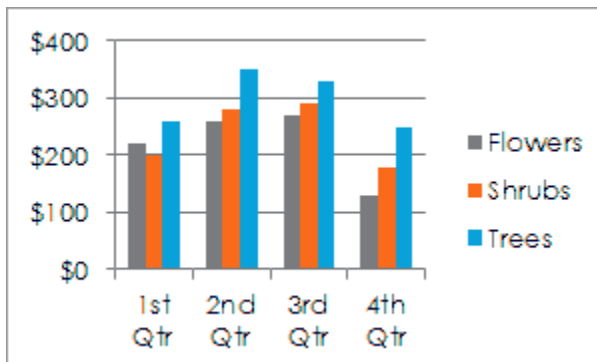
➤ Clustered Column Chart

- Unlike the rest of the column charts, the **clustered column chart** does work well with both single and multiple data series. Graphing a single data series is most effective when there is a large number of time periods (e.g. months, days in a month) as shown below.



➤ Clustered Column Chart (2-D) with Single Data Series

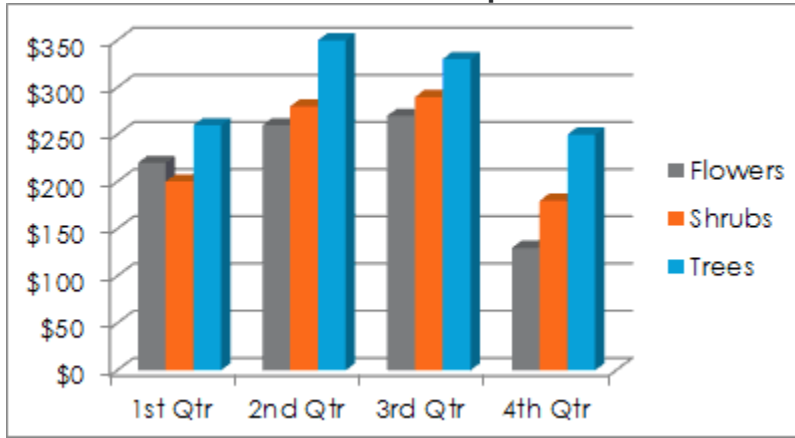
- The column chart works brilliantly with multiple data series. Each series has its own color and values per time period can easily be compared. Our multiple data series described above is graphed in the clustered column chart below.



➤ Clustered Column Chart (2-D)

- Excel places a legend on the chart explaining what data series each color represents. Now let's look at the three-dimensional version of this chart.
- **Clustered Column Chart in 3-D**

- Three-dimensional charts are often more attractive when included in a presentation such as a PowerPoint slide. Here is our data plotted in 3-D.



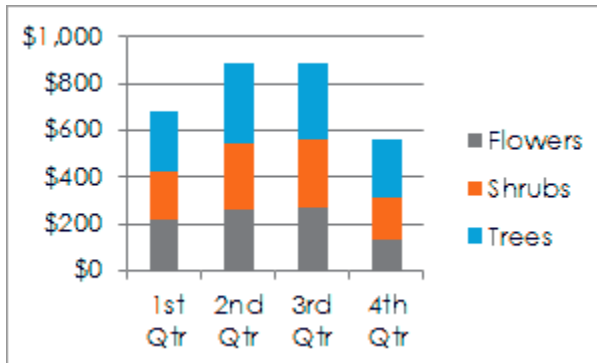
- 3-D Clustered Column Chart

- Though the columns are 3-dimensional, the Z-axis has no values in the clustered column chart.

- [A KeynoteSupport.com Tutorial](http://KeynoteSupport.com)

➤ **Stacked Column Chart**

- In the stacked column chart, the values for each time period or category are stacked on top of each other. As can be seen on our chart's vertical axis number scale, their values are summed as the items are stacked.



Stacked Column Chart (2-

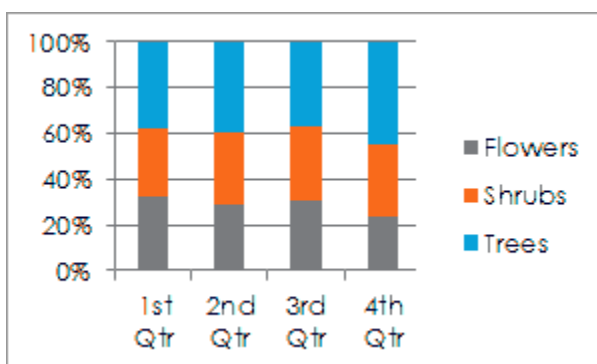
D)

- The stacked column chart shows each value with an appropriately-sized bar segment, but also emphasizes each value's portion of the total value for that time period or category.

The Stacked Column Chart is available in two-dimension as shown, or in 3-D.

100% Stacked Column Chart

In the 100% stacked column chart, the values for each time period are presented as their portion or percentage of 100%.



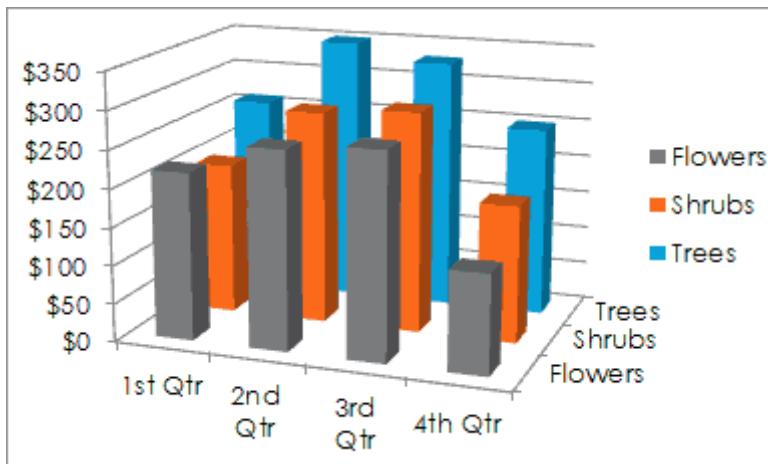
100% Stacked Column

Chart (2-D)

- The 100% stacked column chart shows how a value's contribution to the whole changes over time. In our chart, we can see that "Flower" sales contributed less in 4th quarter than other quarters.
- The 100% Stacked Column Chart is available in both 2-D as shown, or in 3-D.

➤ 3-D Column Chart

The 3-D Column Chart is special because the chart itself is three-dimensional and uses the X, Y, and Z axes. Excel places the vertical bars, in 3-D, on a plane, and the Z-axis (depth) plots the data series.



3-D Column Chart

➤ Line chart:-

What is a Line Chart?

Line Charts, which work especially well with data arranged in columns and rows, show trends over time or other category.

Line charts plot values in horizontal lines. Excel displays each line in a different color.

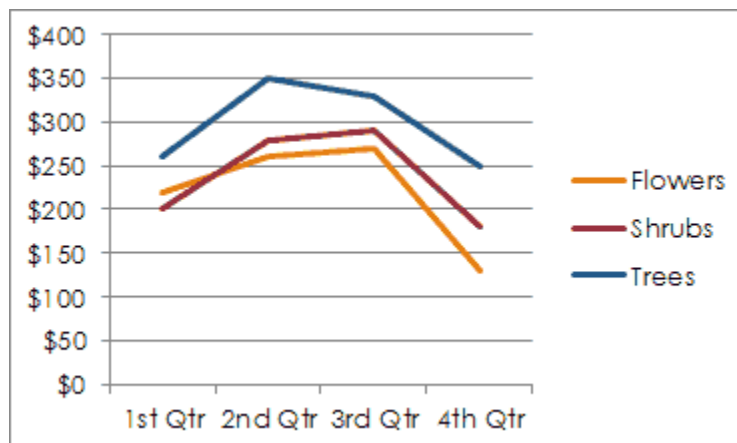
	A	B	C	D	E	F
1		1st Qtr	2nd Qtr	3rd Qtr	4th Qtr	Year
2	Flowers	\$220	\$260	\$270	\$130	\$880
3	Shrubs	\$200	\$280	\$290	\$180	\$950
4	Trees	\$260	\$350	\$330	\$250	\$1,190

Single Data Series

Let's look at some line charts using the data series selected in the above worksheet. Our selection includes multiple data series. When selecting data for a line chart, include row and column headings when possible, but do not include worksheet totals.

2-D Line Chart

In the basic 2-D line chart, Excel will display the numeric values along the vertical axis (Y-axis), while the horizontal axis (X-axis) displays time periods (quarters, years) or other categories evenly scaled.

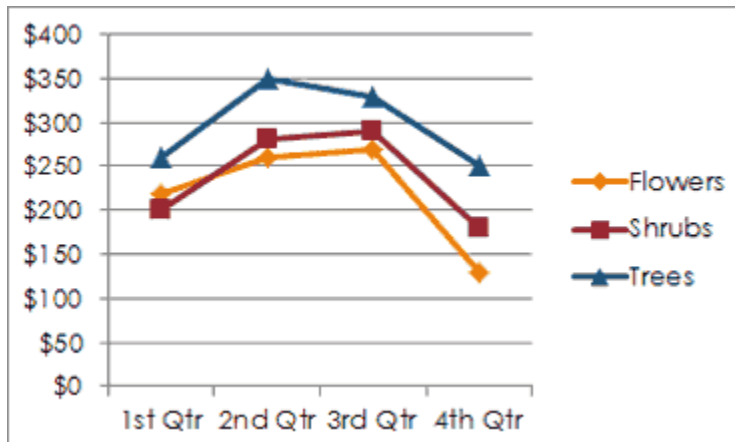


2D line Chart

Line charts can often handle large data series, but can become confusing if the lines overlap too much.

Line Chart with Markers

Markers can be added to locate each data value. Excel chooses from a variety of shapes for each data series' marker.



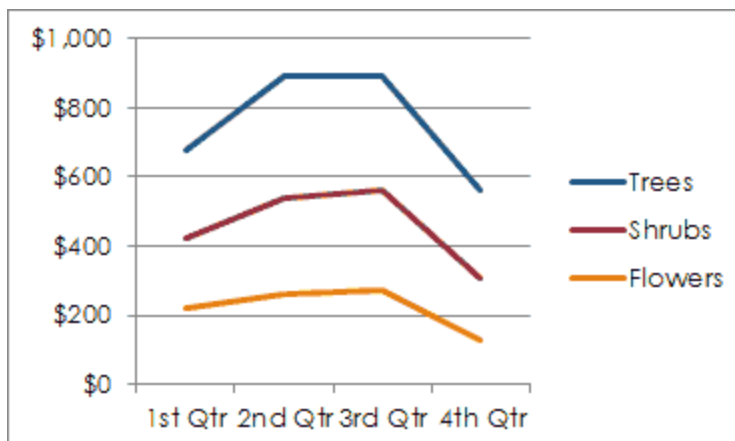
2D Line Chart with Markers

Markers do not work well in line charts crowded with data points as markers overlap each other and trends are obscured.

Stacked Line Chart

The stacked line chart is used to compare trends. In a stacked line chart, the lines plotting the values are stacked on top of each other with the values on the vertical axis cumulative.

Especially in line charts where lines may cross each other, stacking the lines so they are spaced apart allows the trends to be more clearly seen and compared.



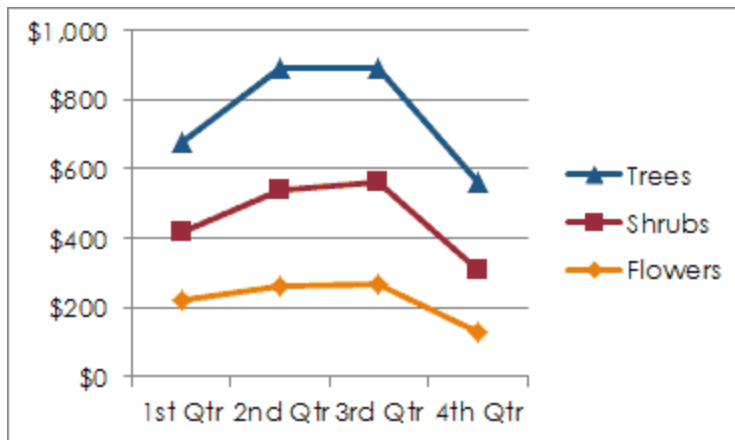
Stacked Line Chart

Excel builds the chart by placing the top row of the data series at the bottom of the chart. Therefore, the lines near the top of the line chart may show a much sharper rise or fall than they would on a 2-D line chart because of the cumulative effect of the stacked chart.

For example, you can see that the line for "Trees" in the stacked chart is different than how it appears on the two-dimensional line chart shown above.

Stacked Line Chart with Markers

The stacked line chart with markers is available in most versions of Microsoft Excel.



Stacked Line Chart with Markers

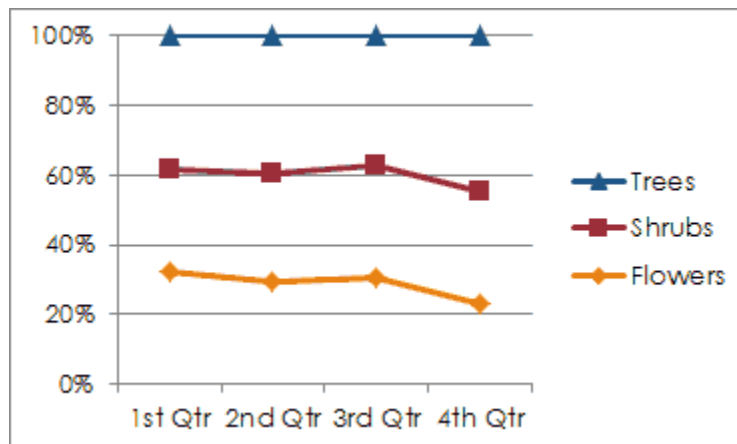
As mentioned, adding markers to any line chart works best only if the markers are not too close to each other.

100% Stacked Line Chart

In a 100% stacked line chart, the lines are scaled so their total is 100%—the vertical axis shows percentages instead of values. This chart type shows the proportion that the data contributes to the trend.

This chart type, shown with markers in the next section, is seldom used as it's difficult for most folks to interpret and understand.

100% Stacked Line Chart with Markers

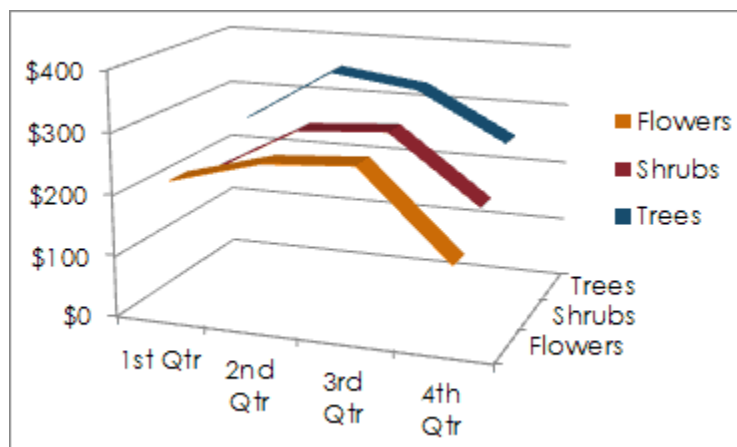


100% Stacked Line Chart with Markers

The 100% stacked line chart with markers is shown above. For each data series, Excel places on the graph one of a variety of shapes, such as squares, circles, and diamonds, for its values.

3-D Line Chart

Another variation of the line chart in Excel is the three-dimensional line chart as seen below.



3-D Line Chart

In the 3-D line chart, Excel displays each data series with a three-dimensional flexible bar or ribbon.

How to Create a LINE Chart in Excel

- Select the cell range to chart. Include row or column heading(s) if possible.
- In the **Charts** section of the ribbon's **Insert tab**, click the small arrow by **Line**.
- From the drop-down menu that displays, select one of the available Line charts.

➤ Pie chart:-

What is a PIE Chart?

A pie chart displays a single data series as proportional slices of a pie, visually showing how much each individual value contribute to the whole. We've outlined our single data series in the worksheet below.

There are a variety of pie charts available in Microsoft Excel. We'll discuss the top variations by charting the five plant categories shown below.

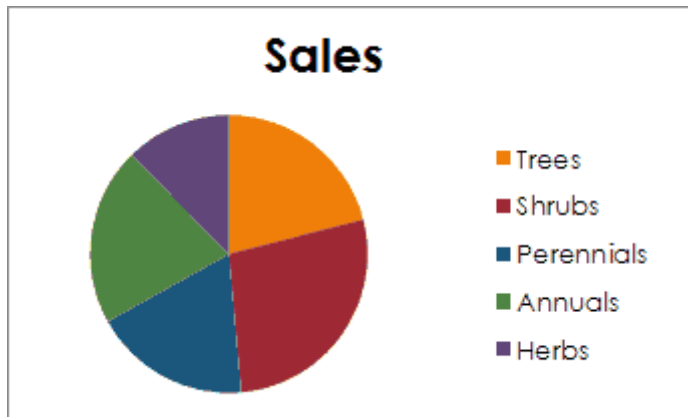
	A	B	C
1		Sales	Units
2	Trees	\$4,890	49
3	Shrubs	\$5,460	95
4	Annuals	\$3,440	267
5	Perennials	\$2,130	95
6	Herbs	\$1,350	168

Excel uses the data heading (series identifier) for the chart title ("Sales" in our chart) and shows the category names in a legend. The pie chart works best when the number of categories is seven or less; and no values are equal to, or less than, zero.

One can [customize the design of the pie chart](#) so either numeric values or their percentages display on top of the slices of pie. Now let's look in detail at the different pie charts available in Microsoft Excel.

The 2-D PIE Chart

The well-known 2-D pie chart is not fancy, but adequately displays and compares the portions of a whole for a small quantity of values. If category headings were also selected, they will display in a legend.



2-D Pie Chart

Legend names can be changed by changing the headings in the worksheet or editing the chart directly in newer versions of Excel. And the entire legend can be moved to the top, bottom, left, right, or top right ("corner" in older versions) of the chart.

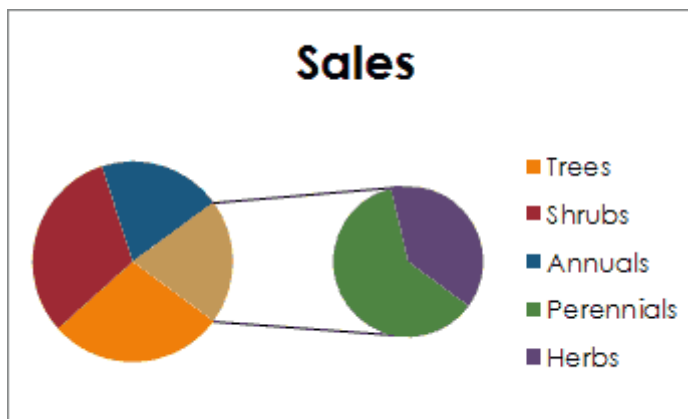
If you don't like the colors of the pie slices, change your worksheet's color theme. Move the chart to the right some, click **Colors** on the **Themes section** of the **Page Layout tab**, and note which theme is outlined in case you want to return to the current theme.

Then hover your cursor over the built-in color themes and watch the colors in your chart change. Click on a theme to select it.

The PIE of PIE Chart

The pie of pie chart makes it easier to view select slices of a pie by enlarging them in a second pie. Connecting lines to the smaller pie show the viewer which portion of the main pie has been enlarged. How does it work?

Modern versions of Excel default to the lowest 3 values in the additional pie. Older versions might use the bottom 2 values. The values to use can be customized.



Pie of Pie Chart

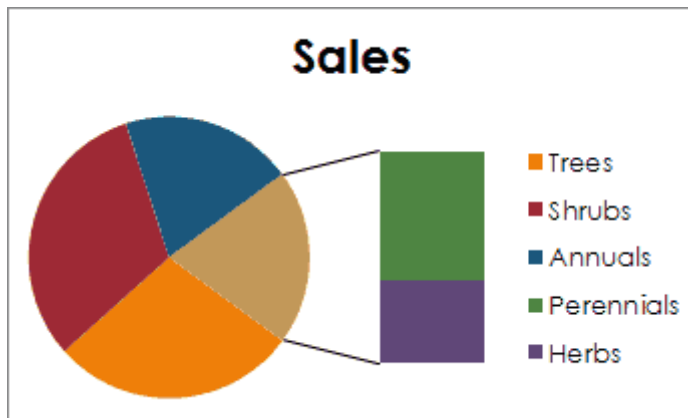
To **add, remove, or change the values used for the secondary pie**, right-click on the secondary pie and click **Format Data Series**.

The **Split Series By** option allows the user to customize how the values are selected (amount, position) and how many values to put inside the secondary pie. Other formatting options are available from the Format Data Series dialog box.

Right-click on the legend to change the font or format the legend's borders, colors, and position.

The BAR of PIE Chart

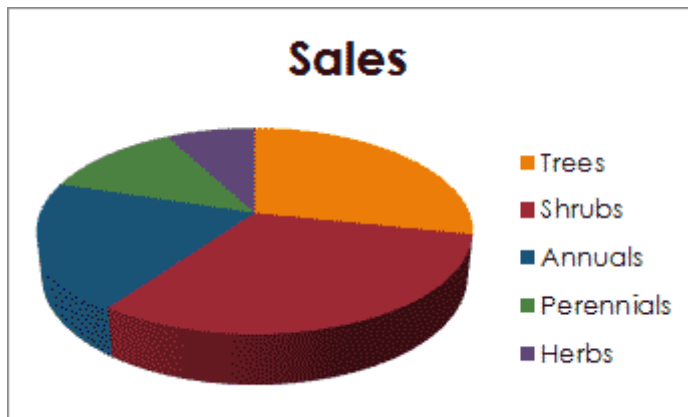
The bar of pie chart is almost identical to the pie of pie, except the data pulled from the main pie is enlarged in a bar instead of a pie.



Bar of Pie Chart

The PIE in 3-D Chart

When creating pie charts for eventual inclusion in a presentation, 3-D pie charts are popular.

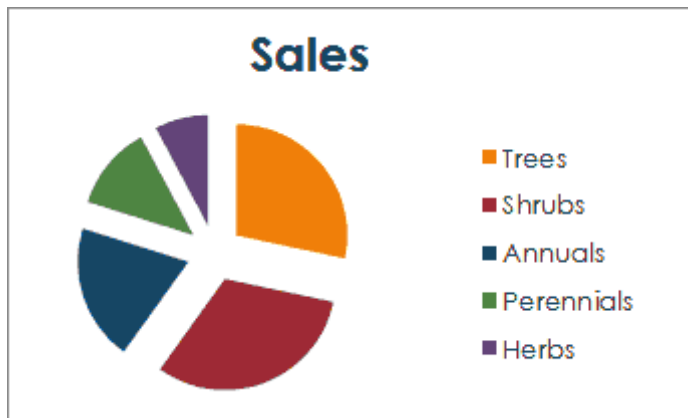


Pie in 3-D

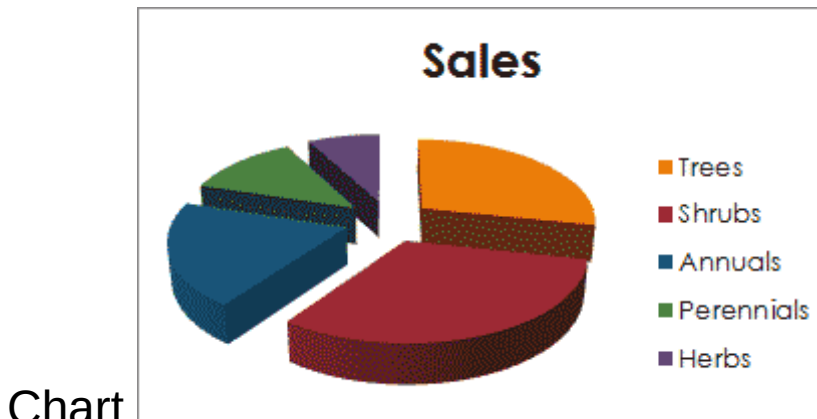
This attractive chart shows the pie in three-dimensions. With six data series, we had to change our worksheet's color theme to one that displayed more unique colors.

The 2-D Exploded PIE and 3-D Exploded PIE Chart

The 2-D and 3-D exploded pie charts are just like the regular 2-D and 3-D pie charts except the pie pieces have been separated as shown below.



2-D Exploded Pie



Chart

Exploded Pie in 3-D

If you don't see an option to create a 2-D exploded pie chart, in most cases you can create a 2-D pie chart manually by double-clicking on a pie piece and dragging it away from the center.

[A KeynoteSupport.com Tutorial](http://AKeynoteSupport.com)

How to Create a PIE Chart

To create a Pie chart in Excel, follow these steps:

- Highlight a single data series including column and row headers if possible.
- Click the **Insert tab**.
- In the Charts section of the tab, click **Pie** and select one of the pie charts offered.

➤ Bar chart:-

What is a BAR Chart?

A bar chart visually illustrates data over certain time periods using horizontal bars or other shapes. Numeric values are typically plotted on the horizontal axis.

The Bar Chart looks like a [Column Chart](#) lying on its side, and tends to display and compare a large number of data series better than other types of charts.

Simple BAR Chart: Single Data Series

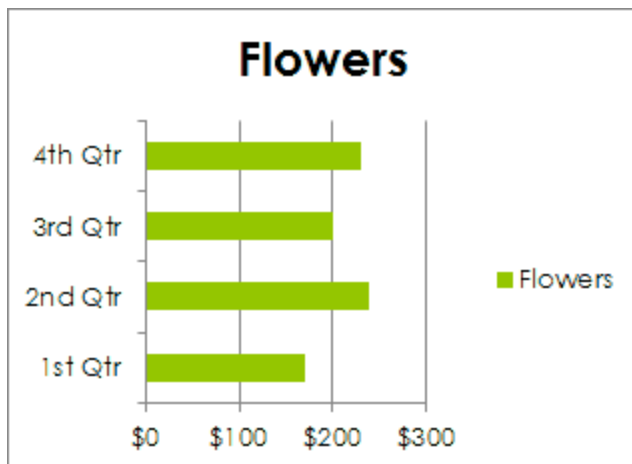
A simple bar chart displays one category over a period of time. This is called graphing a **single data series**. A data series is a row or column of data.

	A	B	C	D	E	F
1		1st Qtr	2nd Qtr	3rd Qtr	4th Qtr	Year
2	Flowers	\$170	\$240	\$200	\$230	\$840
3	Shrubs	\$220	\$280	\$250	\$290	\$1,040
4	Trees	\$260	\$340	\$200	\$320	\$1,120

In our worksheet above, we've outlined our single data series consisting of the values representing "Flowers" revenue for each of the four quarters of the year.

Notice that we didn't include the Year totals in our data series, just the individual quarters for comparison. But we did include the row and column headings as we want them to display on our chart.

We've charted below, in a 2D Clustered Bar Chart, the single data series shown in the above worksheet. We can visually see how the revenue compares for the four quarters.



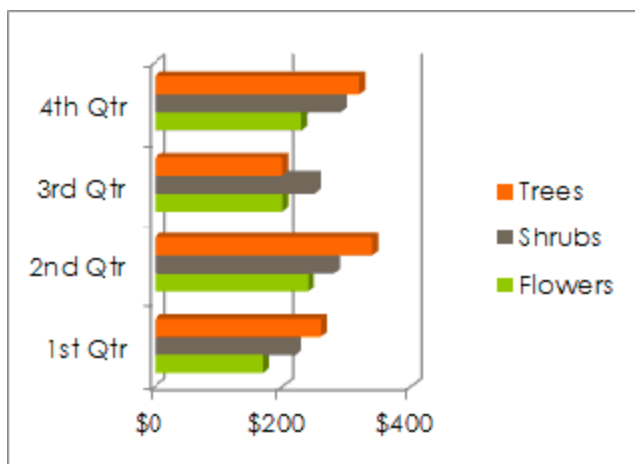
When charting a single data series, all the bars are the same color. Also, the two-dimensional chart is flat and not very appealing, but we'll look at design options later.

Complex BAR Chart: Multiple Data Series

	A	B	C	D	E	F
1		1st Qtr	2nd Qtr	3rd Qtr	4th Qtr	Year
2	Flowers	\$170	\$240	\$200	\$230	\$840
3	Shrubs	\$220	\$280	\$250	\$290	\$1,040
4	Trees	\$260	\$340	\$200	\$320	\$1,120

Now we'll chart **multiple data series**. We'll chart Flowers, Shrubs, and Trees over the four quarters of the year. Our selection is outlined in the above worksheet.

We've created a 3D Cluster Bar Chart shown below. Each item or category is presented in a different color as explained by the chart legend. The colors depend on the Color Theme chosen for the Excel workbook. (Themes are found on Excel's **Page Layout** tab.)

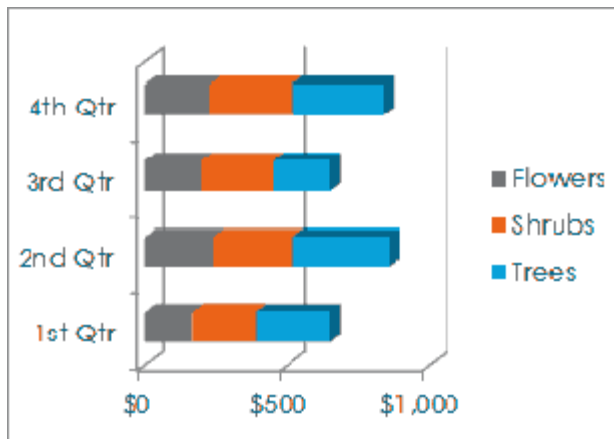


If the number of categories and time periods are about equal, switch them around on the chart for a different look at the data. To do so, click on the chart, click the **Design** tab on the ribbon, and click **Switch Row/Column**.

[A KeynoteSupport.com Tutorial](http://KeynoteSupport.com)

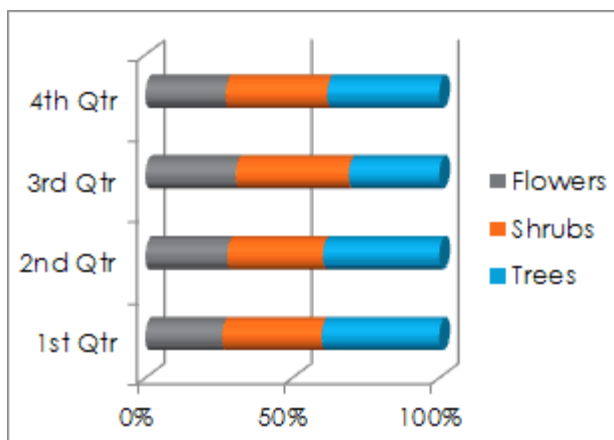
Other Types of BAR Charts

Excel offers several kinds of bar charts. The most popular, the Clustered Bar Chart, was shown above in both 2D and 3D formats. Two other sub-types are available.



Stacked Bar Chart in 3-D

The Stacked Bar Chart compares the contribution that each **value** make to the total. Our multiple data series is plotted in the above chart. Notice that for each quarter, the three categories are horizontally stacked atop each other.



100% Stacked Bar Chart in

3-D with Cylinders

The 100% Stacked Bar Chart, shown above, compares the **percentage** that each value contributes to the total, with the horizontal axis ranging from 0 to 100%.

Both the Stacked Bar Chart and 100% Stacked Bar Chart are available in two-dimension and three-dimension, and both are meaningful only when graphing multiple data series.



Bar, Cylinder, Cone, and Pyramid

In newer versions of Excel, cylinders, cones, or pyramids, shown in the above image, can be used instead of bars for the three types of bar charts.

How to Create a BAR Chart in Excel

Follow these steps to create a bar chart.

- Select a cell range you wish to plot. Include row or column heading(s) if possible.
- On the **Insert tab** of the Excel ribbon, find the **Charts section** and click the small arrow by **Bar**.
- From the drop-down menu that displays, select one of the available bar charts.

➤ Scatter chart:-

What is a Scatter Chart?

Unlike most other charts which focus on amounts or trends, **scatter charts show the relationship between two variables**. Scatter charts plot number pairs, similar to how we plot ordered pairs in the Cartesian coordinate system (X,Y). In fact, the scatter chart is also called an **XY chart**.

The relationship between two points on a scatter chart is called their correlation. If the data points on a scatter chart hug a **trendline** (see last section) progressing upward, this may indicate a **positive correlation** where both variables increase.

Conversely, if values hug a trendline progressing downward, this may indicate a **negative correlation**. And data points randomly dispersed suggest a lack of correlation.

In Excel, scatter charts are available with markers only, smooth or straight lines only, or a combination of the two.

AKeynoteSupport.com Tutorial

Scatter Chart with Only Markers

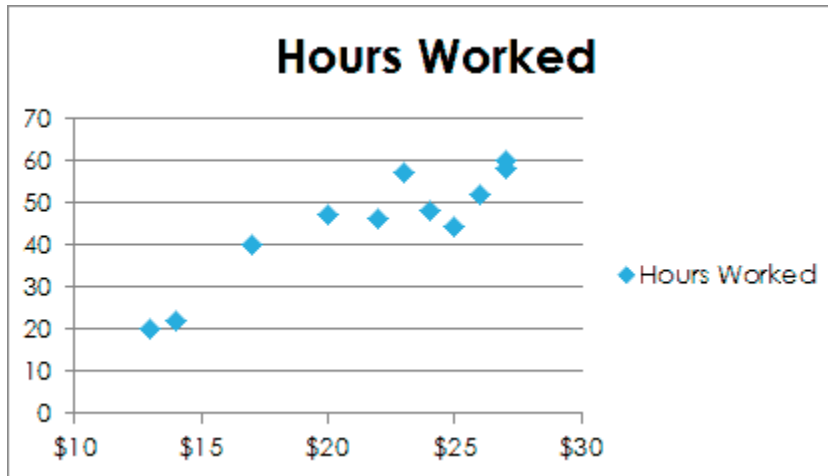
The most frequently used scatter chart is one with markers only. Each point on the scatter chart, represented by

a **marker**, is the intersection of a pair of numbers from two columns or two rows.

The number pairs in our worksheet are from two columns labeled "Hourly Wage" and "Hours Worked." To create our scatter chart, we selected column headings and the data, but NOT the row headings.

	A	B	C
1	Age	Hourly Wage	Hours Worked
2	18-22	\$13	20
3	23-27	\$20	47
4	28-32	\$22	46
5	33-37	\$25	44
6	38-42	\$26	52
7	43-47	\$27	58
8	48-52	\$27	60
9	53-57	\$23	57
10	58-62	\$24	48
11	63-67	\$17	40
12	68-72	\$14	22

The values in the left column (or top row if using row data), are plotted on the horizontal axis (X-axis) **in numerical order**. The values in the right column (or bottom row if using row data) are plotted on the vertical axis, and its header provides the legend and chart title.



In our scatter chart, the first data point is where \$13 and 20 intersect. The data pair (13,20) is first because 13 is the smallest value in "Hourly Wage" - the column plotted on the horizontal axis. The second data point is at (14,22) and so on.

Does our scatter chart show a pattern with our data? We'd say there is a mostly positive correlation between "Hours Worked" and "Hourly Wage." Now let's look at some other types of scatter charts available in Excel.

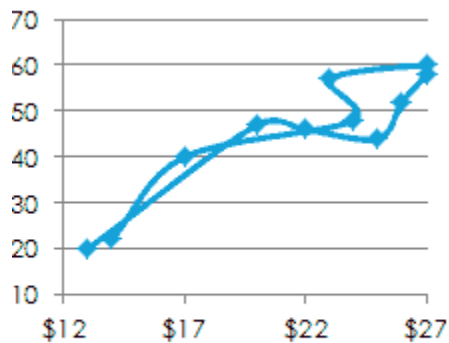
[A KeynoteSupport.com](http://AKeynoteSupport.com) Tutorial

Other Scatter Charts in Microsoft Excel

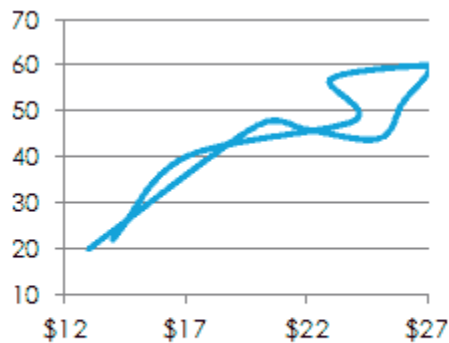
Additional scatter charts available in Excel include:

- Scatter with Smooth Lines and Markers
- Scatter with Smooth Lines
- Scatter with Straight Lines and Markers
- Scatter with Straight Lines

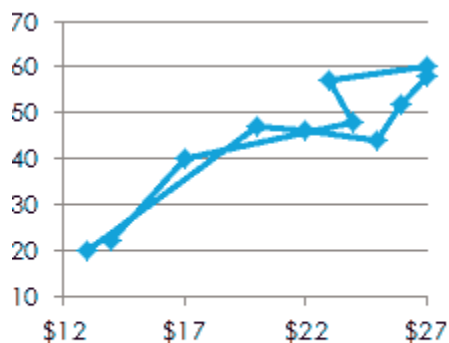
These four scatter charts below show the same data pairs as in our above example. But in our scatter charts, the line crosses itself and zig-zags. (Most tutorials show a nice smooth line in their example!) If this is a concern, see [Microsoft Excel: The Scatter Chart with Lines](#) for a discussion of why this happens and how to fix.



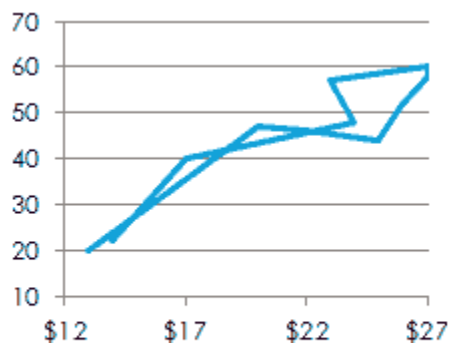
Scatter chart with smooth lines and markers



Scatter chart with smooth lines



Scatter chart with straight lines and markers



Scatter chart with straight lines

In newer versions of Excel, the scatter chart section offers the **Bubble Chart** and **3-D Bubble Chart**. In older versions of Excel, these two charts are located under **Other Charts**.

[A KeynoteSupport.com Tutorial](#)

How to Create a Scatter Chart in Microsoft Excel

Follow these steps to create a scatter chart.

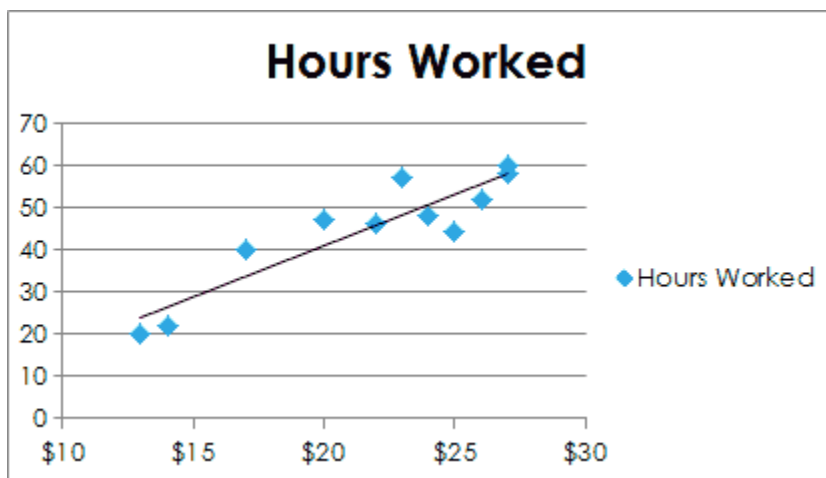
- Select a cell range you wish to plot consisting of an equal number of cells from two columns or two rows.
- On the **Insert tab** of the Excel ribbon, find the **Charts section** and click the small arrow by **Scatter**.
- From the drop-down menu that displays, select a type of scatter chart.

If using an older version of Excel, visit [Beginner's Guide to Excel Charts](#) and follow the steps in the section titled "Create a Chart in Excel 2003, 2000, and 98."

Important Customization Options

► How to Add a Trendline

Adding a **trendline** to a scatter chart helps show if there is a correlation or pattern between the variables. An approximate equal number of data points reside above and below the trendline.



To draw a trendline on a scatter chart in Microsoft Excel, right-click on any chart data point (marker) and click **Add Trendline** from the menu.

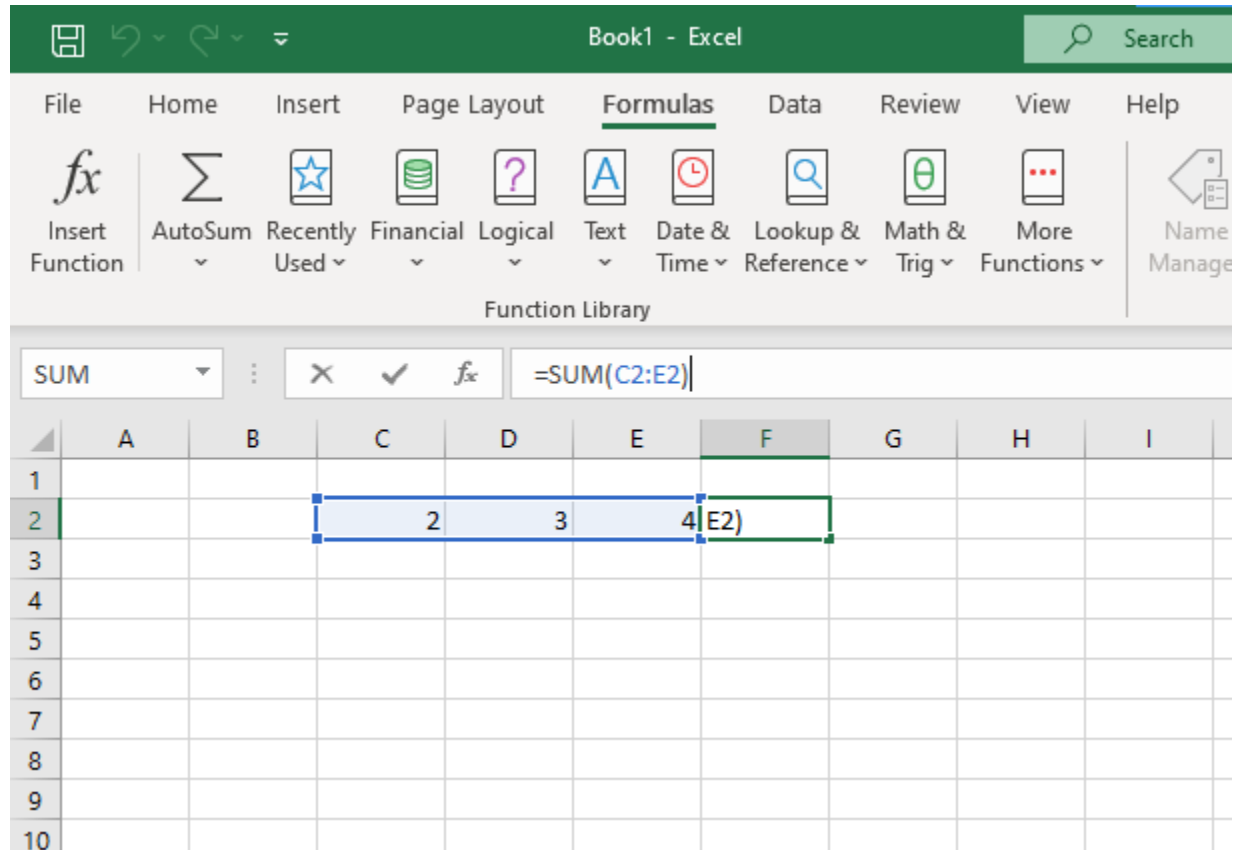
❖ **Mathematical function:-**

- A very important feature in excel is the formula. It is used to calculate value based on what is cells perform operation on a cell content fetch value after an operation based on your search criteria and much more.
- Mathematical formulas in excel are used to perform various arithmetic operations like sum, average, count, max, min etc.. here is all list of most frequently used mathematical formulas in excel.
- **SUM:** This function is used to adds all the values with a cell range.
- **Syntax:** =sum(cell address : cell address);
- **Example :** sum(C1:C2)=15

Here in the example below we will create a basic function to calculate the sum of working hours generates in a day.

- Select the cell where you want to put the formula type the equals sign(=) and write the desired function name or choose the function from the suggested function list . Here in the example below we write the sum function.

- Now press enter key to see the result or press ctrl + enter key to stay in the formula cell.
- **Sum=(numer1,numer2,numer3)**
- **Example: =sum(4,5,6)=15**



- **Sumif();** :- Here in the example below we will create a basic function to calculate the sum of working hours generates in a day only for female employees.
- **Syntax:- sum if(range, criteria);**
Type equals sign(=) and write the desired function in the cell.
- press enter key to see the result or press ctrl + enter key to stay in the formula cell.

- **Average :-** The **AVERAGE** function is a premade function in Excel, which calculates the average (arithmetic mean).

It is typed **=AVERAGE**

It adds the range and divides it by the number of observations.

Example:

The average of (2, 3, 4) is 3.

3 observations (2, 3 and 4)

The sum of the observations ($2 + 3 + 4 = 9$)

($9 / 3 = 3$)

The average is 3

How to use the **=AVERAGE** function, step by step:

1. Select a cell (**F2**)
2. Type **=AVERAGE**
3. Double click the AVERAGE command
4. Select a range (**B2:E2**)
5. Hit enter
6. Next, Fill **F2:F7**

F2						f_x	=AVERAGE(			
	A	B	C	D	E	F	G	H		
1	Trainer	Pokeball	Great ball	Ultraball	Master ball	Average				
2	Iva	10	4	1	1	=AVERAGE(				
3	Liam	12	3	0	1	AVERAGE (number1; [number2]; ...)				
4	Jenny	15	1	3	1					
5	Iben	4	2	6	0					
6	Adora	10	4	1	1					
7	Kasper	9	2	1	0					

1. Select a cell
2. Type **=COUNT**
3. Double click the **COUNT** command
4. Select a range
5. Hit enter
6. Hit enter

	A	B	C	D	E	F	G	H	I
7	Cewgong	Water	Ice	475					
8	Cloyster	Water	Ice	525					
9	Onix	Rock	Ground	385					
10	Dragonair	Dragon		420					
11	Pidgeotto	Normal	Flying	349					
12	Rattata	Normal		253					
13	Beedrill	Bug	Poison	395					
14	Doduo	Normal	Flying	310					
15	Kingler	Water		475					
16	Nidoqueen	Poison	Ground	505					
17	Hitmonchan	Fighting		455					
18	Charmeleon	Fire		405					
19	Arbok	Poison		438					
20	Gastly	Ghost	Poison	310					
21	Magikarp	Water		200					
22									
23				=count					
24				 COUNT	Counts the number of cells in a range that contain numbers				
25									

Note: The **=COUNT** function only counts cells with numbers in a range.

	A	B	C	D	E	F
1	Name	Type 1	Type 2	Total stats		
2	Mankey	Fighting		305		
3	Poliwrath	Water	Fighting	510		
4	Victreebel	Grass	Poison	490		
5	Tentacool	Water	Poison	335		
6	Magnetron	Electric	Steel	465		
7	Dewgong	Water	Ice	475		
8	Cloyster	Water	Ice	525		
9	Onix	Rock	Ground	385		
10	Dragonair	Dragon		420		
11	Pidgeotto	Normal	Flying	349		
12	Rattata	Normal		253		
13	Beedrill	Bug	Poison	395		
14	Doduo	Normal	Flying	310		
15	Kingler	Water		475		
16	Nidoqueen	Poison	Ground	505		
17	Hitmonchan	Fighting		455		
18	Charmeleon	Fire		405		
19	Arbok	Poison		438		
20	Gastly	Ghost	Poison	310		
21	Magikarp	Water		200		
22						
23				=COUNT(D2:D21)		
24				COUNT (value1; [value2]; ...)		
25						

	A	B	C	D
1	Name	Type 1	Type 2	Total stats
2	Mankey	Fighting		305
3	Poliwrath	Water	Fighting	510
4	Victreebel	Grass	Poison	490
5	Tentacool	Water	Poison	335
6	Magnetron	Electric	Steel	465
7	Dewgong	Water	Ice	475
8	Cloyster	Water	Ice	525
9	Onix	Rock	Ground	385
10	Dragonair	Dragon		420
11	Pidgeotto	Normal	Flying	349
12	Rattata	Normal		253
13	Beedrill	Bug	Poison	395
14	Doduo	Normal	Flying	310
15	Kingler	Water		475
16	Nidoqueen	Poison	Ground	505
17	Hitmonchan	Fighting		455
18	Charmeleon	Fire		405
19	Arbok	Poison		438
20	Gastly	Ghost	Poison	310
21	Magikarp	Water		200
22				
23				20
24				

That's it! The **=COUNT** function successfully counted 20 cells with numbers.

➤ Rand:-

The **RAND** function is used to generate random numbers.

It is typed **=RAND**

Rand can be used to generate any random number.

You can define limits, create random data sets and much more.

To use the default **RAND** function, write:

=RAND()

To use the **RAND** function to receive a random number up to a certain value, write:

=RAND()*value

To use the **RAND** function to receive a whole number, write:

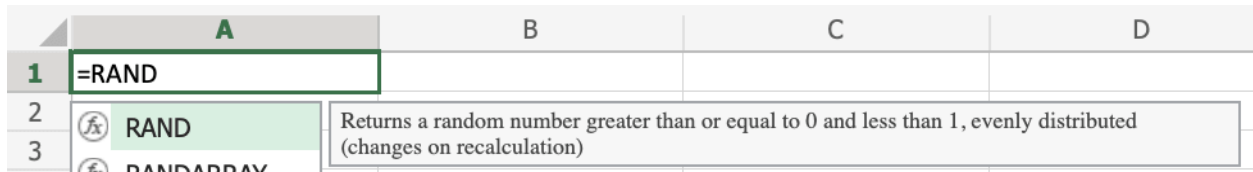
=INT(RAND())

How to Use RAND Function

To receive a random number, use **RAND**.

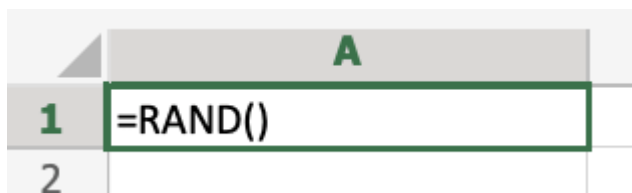
Step 1) Start the RAND function

1. Select a cell **A1**
2. Type **=RAND**
3. Double click the **RAND** command

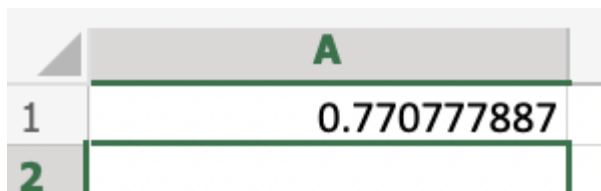


Step 2) Close the function

1. Close the function with **()**
2. Hit enter



The random number has been generated!



How to Use RAND Function With a Defined Limit Value.

1. Define the limit (***10**)
2. Hit enter

	A
1	=RAND()*10
2	

A random number between 0 and 10 has been generated!

	A
1	7.931180668
2	

How to Receive a Whole Number Using The RAND Function.

1. Add (**INT**) before the **RAND** function
2. Hit enter

	A
1	=INT(RAND()*10)
2	

A random whole number between 0 and 10 has been generated!

	A
1	5
2	

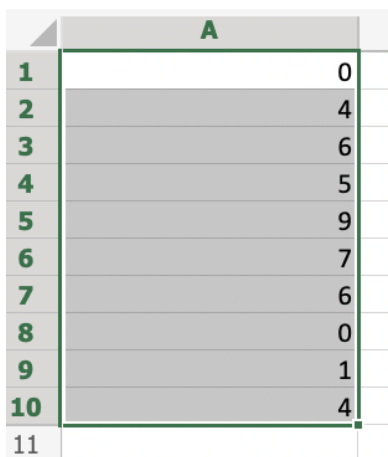
How to Generate Multiple Random Numbers.

1. Drag the bottom right corner of the cell down to generate more numbers.



	A
1	5
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	

10 random numbers were generated in the **A** column!



	A
1	0
2	4
3	6
4	5
5	9
6	7
7	6
8	0
9	1
10	4
11	

❖ Logical function:-

- If function:-

The if function is the key logical function used for decision making ,If takes the formula.

=if(condition,true,false);

For example, you could use the following formula .

=if(B2>400"high","low); where B2>400 is the condition tested high is the test to display. If is greater than 400(true).

"low" is the text to display if B2 is less than or equal to 400 (false).

- **AND function:-**

The AND function is used to compare more than one collection. It returns true only if all of the conditions are met and takes the formula.

=AND(B2>400,C2<400);

Where B2>400 is the first condition being tested C2<400 is the second condition being tested. This all only return the result true if the value in cell B2 is greater than 400 and the value in cell C2 is less than 400 in all other situation the result will be false.

- **OR function:-**

The OR function is also used to combine more than one condition. It is return true is any of the condition one be are met and to the format.

=OR(condition, condition....);

For example you could use the following formula.

=OR(B2>400, C2<300) ;

Where B2>400 is the first condition being tested

C2<300 is the second condition being tested

This will return the true if either if the value in cell B2 is greater than 400 or the value in cell C2 is less than 300 the result will be false only in neither of condition is met.

❖ **Date function:-**

The excel date function creates a valid date individual years, month, and day components. The date function is use full for assembling dates that need to change dynamically base on other values in a worksheet.

- **Purpose:-** create a date with year, month, and day.
- **Return value:-** A valid excel date.
- **Syntax:-** =Date(year,month,day)
- **Arguments:-** year= numbers for year
- **Version:-** Excel 2003
- **Example 1 hard-coded number**

For example you can use the DATE function to create the dates January 1, 1999 and June 2010 with the following syntax.

Following syntax:

=DATE(1999,1,1)

=DATE(2010,6,1)

- **Example 2 cell reference:-**

The date function is use full for assembling dates that need to change dynamically based on other inputs in a worksheet, for example with 2018 in cell A1, the formula below returns the date April 15,2018.

=DATE(A1,4,15)11 April 15, 2018

- **Example 3 – with SUMIFS COUNTES**

The date function can be used to supply dates as inputs to other function like SUMIFS or COUNTIFS since you can easily assemble a date using year.

=COUNTF(range,">"& date(A1,B1,C1)

- **Example 4- First day of current year**

To return the fast day of the current year you can use the date function like this:

=DATE(yea(today()),1,1); First of year